

Green Bonds EDA

Noah Leibowitz & Siya Motwani

2026-03-09

```
# Load in all libraries
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v forcats 1.0.0 v readr 2.1.5
```

```
## v ggplot2 3.5.1 v stringr 1.5.1
```

```
## v lubridate 1.9.4 v tibble 3.2.1
```

```
## v purrr 1.0.2 v tidyr 1.3.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag() masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
```

```
library(ggplot2)
```

```
library(maps)
```

```
## Warning: package 'maps' was built under R version 4.4.3
```

```
##
```

```
## Attaching package: 'maps'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## map
```

```
library(tidyr)
```

```
# Solar data sets
solar_2024 = read_excel("Solar.xlsx", sheet = "2024")
solar_2023 = read_excel("Solar.xlsx", sheet = "2023")
solar_2022 = read_excel("Solar.xlsx", sheet = "2022")
solar_2021 = read_excel("Solar.xlsx", sheet = "2021")
solar_2020 = read_excel("Solar.xlsx", sheet = "2020")
solar_2019 = read_excel("Solar.xlsx", sheet = "2019")
solar_2018 = read_excel("Solar.xlsx", sheet = "2018")
solar_2017 = read_excel("Solar.xlsx", sheet = "2017")
solar_2016 = read_excel("Solar.xlsx", sheet = "2016")
solar_2015 = read_excel("Solar.xlsx", sheet = "2015")
solar_2014 = read_excel("Solar.xlsx", sheet = "2014")
solar_2013 = read_excel("Solar.xlsx", sheet = "2013")
```

```
# Wind data sets
wind_2024 = read_excel("Wind.xlsx", sheet = "2024")
wind_2023 = read_excel("Wind.xlsx", sheet = "2023")
wind_2022 = read_excel("Wind.xlsx", sheet = "2022")
wind_2021 = read_excel("Wind.xlsx", sheet = "2021")
wind_2020 = read_excel("Wind.xlsx", sheet = "2020")
wind_2019 = read_excel("Wind.xlsx", sheet = "2019")
wind_2018 = read_excel("Wind.xlsx", sheet = "2018")
wind_2017 = read_excel("Wind.xlsx", sheet = "2017")
wind_2016 = read_excel("Wind.xlsx", sheet = "2016")
wind_2015 = read_excel("Wind.xlsx", sheet = "2015")
wind_2014 = read_excel("Wind.xlsx", sheet = "2014")
wind_2013 = read_excel("Wind.xlsx", sheet = "2013")
```

```
# Green Bond Issuance and RPS data set
GBI = read_excel("GB_RPS Hard Code.xlsx")
```

```
# Vector of years
years <- 2024:2013
print("hello")
```

```
## [1] "hello"
```

```
# Initialize empty list to store per-year state totals
solar_totals <- list()

for (yr in years) {
  # Get the dataframe for that year using get()
  df <- get(paste0("solar_", yr))

  # Sum capacity per state
  solar_totals[[as.character(yr)]] <- df %>%
    group_by(State) %>%
    summarise(TotalCapacity = sum(`Nameplate Capacity (MW)`, na.rm = TRUE))
}
```

```

solar_all_years <- bind_rows(
  lapply(names(solar_totals), function(yr) {
    solar_totals[[yr]] %>% mutate(Year = as.numeric(yr))
  })
)

# Initialize empty list to store per-year state totals for wind
wind_totals <- list()

for (yr in years) {
  # Get the dataframe for that year dynamically
  df <- get(paste0("wind_", yr))

  # Sum capacity per state
  wind_totals[[as.character(yr)]] <- df %>%
    group_by(State) %>%
    summarise(TotalCapacity = sum(`Nameplate Capacity (MW)`, na.rm = TRUE))
}

wind_all_years <- bind_rows(
  lapply(names(wind_totals), function(yr) {
    wind_totals[[yr]] %>% mutate(Year = as.numeric(yr))
  })
)

```

```

solar_mean <- bind_rows(
  lapply(names(solar_totals), function(yr) {
    solar_totals[[yr]] %>% mutate(Year = as.numeric(yr))
  })
) %>%
  group_by(State) %>%
  summarise(MeanSolar = mean(TotalCapacity, na.rm = TRUE))

wind_mean <- bind_rows(
  lapply(names(wind_totals), function(yr) {
    wind_totals[[yr]] %>% mutate(Year = as.numeric(yr))
  })
) %>%
  group_by(State) %>%
  summarise(MeanWind = mean(TotalCapacity, na.rm = TRUE))

# Merge solar and wind means
renewables_mean <- full_join(solar_mean, wind_mean, by = "State") %>%
  mutate(MeanRenewable = MeanSolar + MeanWind)

us_states <- map_data("state")

# Make sure state names match: lowercase
renewables_mean$region <- tolower(state.abb[match(renewables_mean$State, state.abb)])

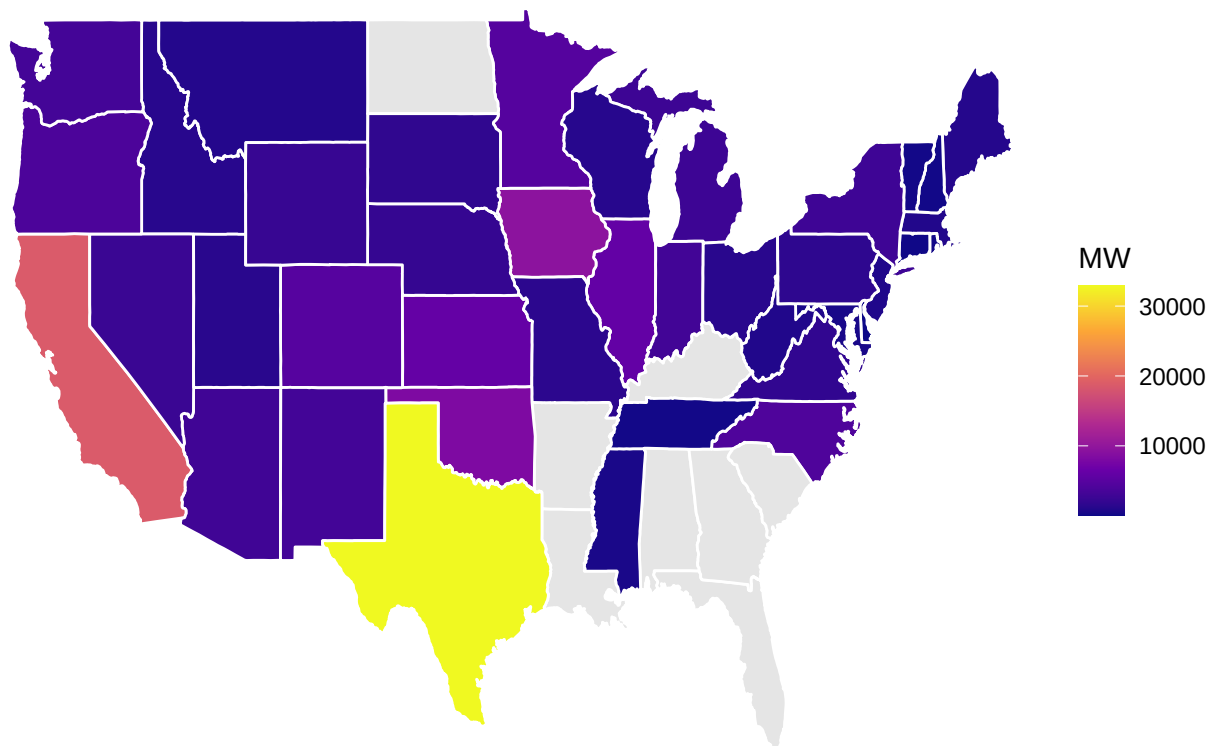
# Convert state abbreviations to lowercase full names
renewables_mean$region <- tolower(state.name[match(renewables_mean$State, state.abb)])

```

```
us_map_data <- left_join(us_states, renewables_mean, by = "region")

ggplot(us_map_data, aes(x = long, y = lat, group = group, fill = MeanRenewable)) +
  geom_polygon(color = "white") +
  scale_fill_viridis_c(option = "C", na.value = "grey90") +
  labs(
    title = "Mean Renewable Energy Capacity by State (MW)",
    fill = "MW"
  ) +
  theme_void() +
  theme(legend.position = "right")
```

Mean Renewable Energy Capacity by State (MW)



```
# Map data
us_states <- map_data("state")

# Merge map with solar and wind means separately
solar_map_data <- left_join(us_states, renewables_mean %>% select(region, MeanSolar), by = "region")
wind_map_data <- left_join(us_states, renewables_mean %>% select(region, MeanWind), by = "region")

solar = ggplot(solar_map_data, aes(x = long, y = lat, group = group, fill = MeanSolar)) +
  geom_polygon(color = "white") +
  scale_fill_gradientn(
    colors = c(
      "#E6D3B3",
      "#D49A73",
```

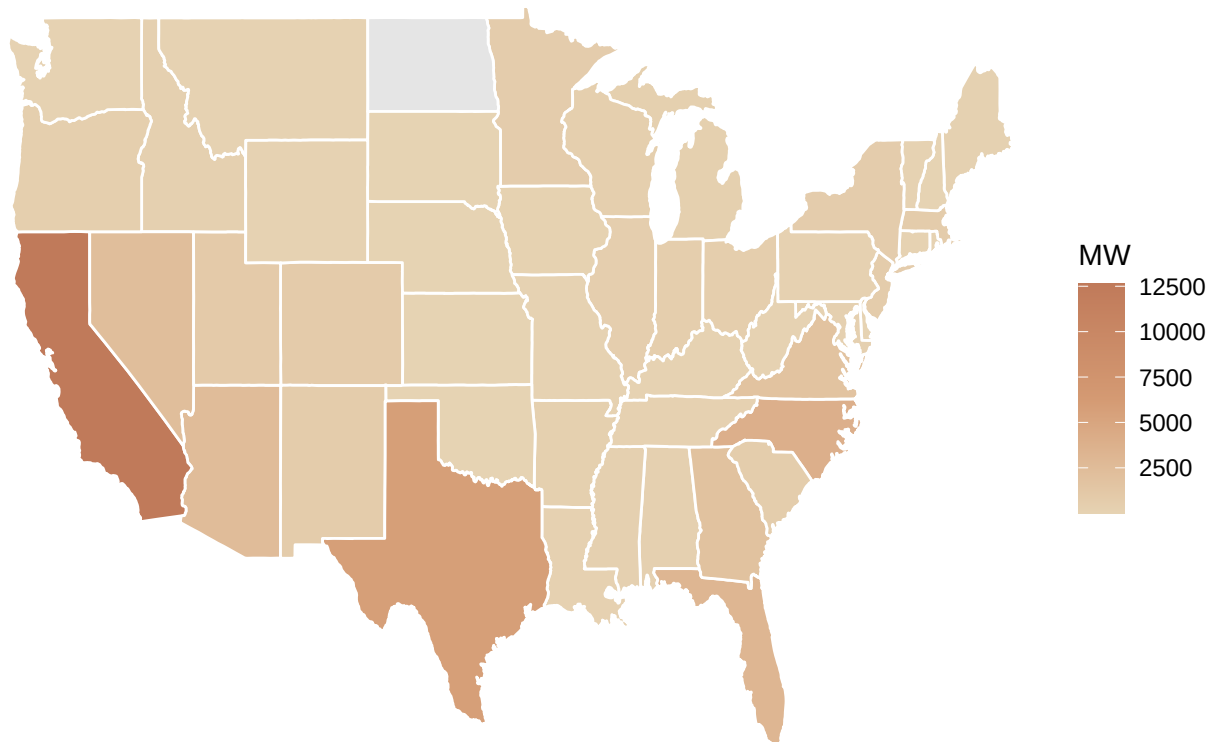
```

    "#C07A5A"
  ),
  na.value = "grey90"
) +
labs(
  title = "Mean Solar Capacity by State (MW)",
  fill = "MW"
) +
theme_void() +
theme(
  legend.position = "right",
  plot.background = element_rect(fill = "white", color = NA),
  panel.background = element_rect(fill = "white", color = NA)
)

print(solar)

```

Mean Solar Capacity by State (MW)



```

ggsave("solar.png", plot = solar, width = 6, height = 4, dpi = 300)

```

```

wind = ggplot(wind_map_data, aes(x = long, y = lat, group = group, fill = MeanWind)) +
  geom_polygon(color = "white") +
  scale_fill_gradientn(
    colors = c(
      "#E6D3B3",
      "#D49A73",

```

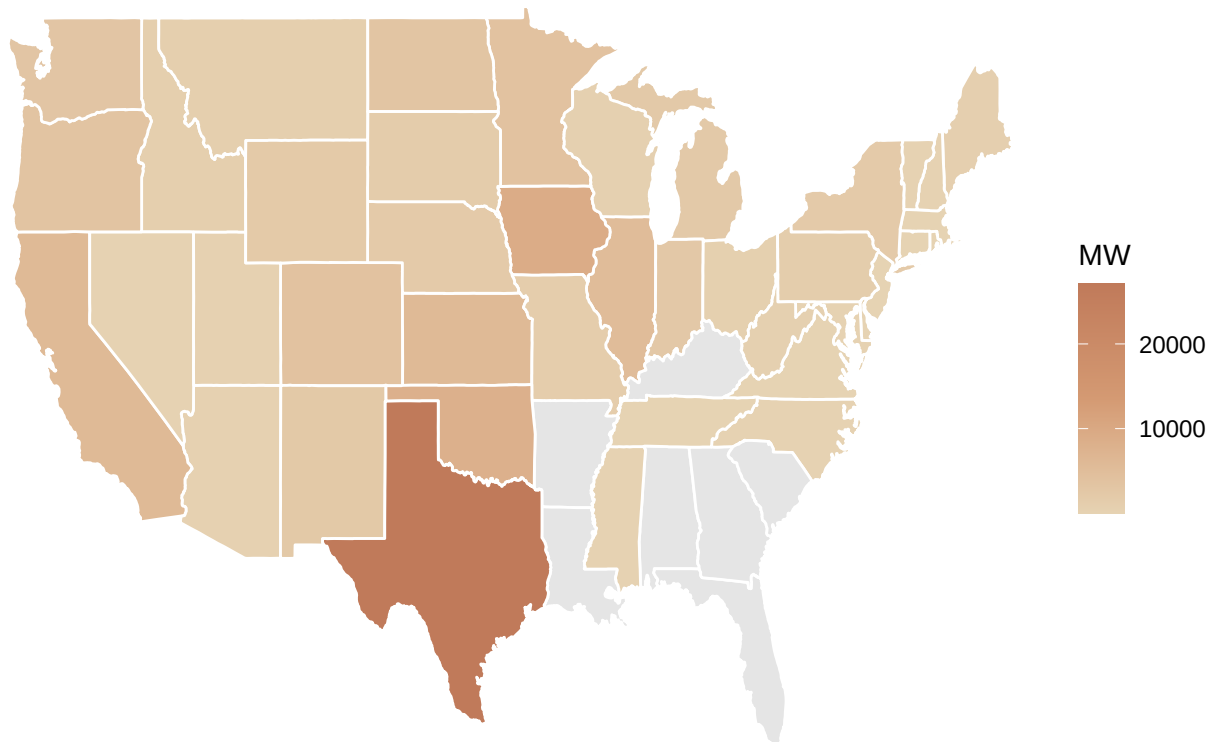
```

    "#C07A5A"
  ),
  na.value = "grey90"
) +
labs(
  title = "Mean Wind Capacity by State (MW)",
  fill = "MW"
) +
theme_void() +
theme(
  legend.position = "right",
  plot.background = element_rect(fill = "white", color = NA),
  panel.background = element_rect(fill = "white", color = NA)
)

print(wind)

```

Mean Wind Capacity by State (MW)



```
ggsave("wind.png", plot = wind, width = 6, height = 4, dpi = 300)
```

```

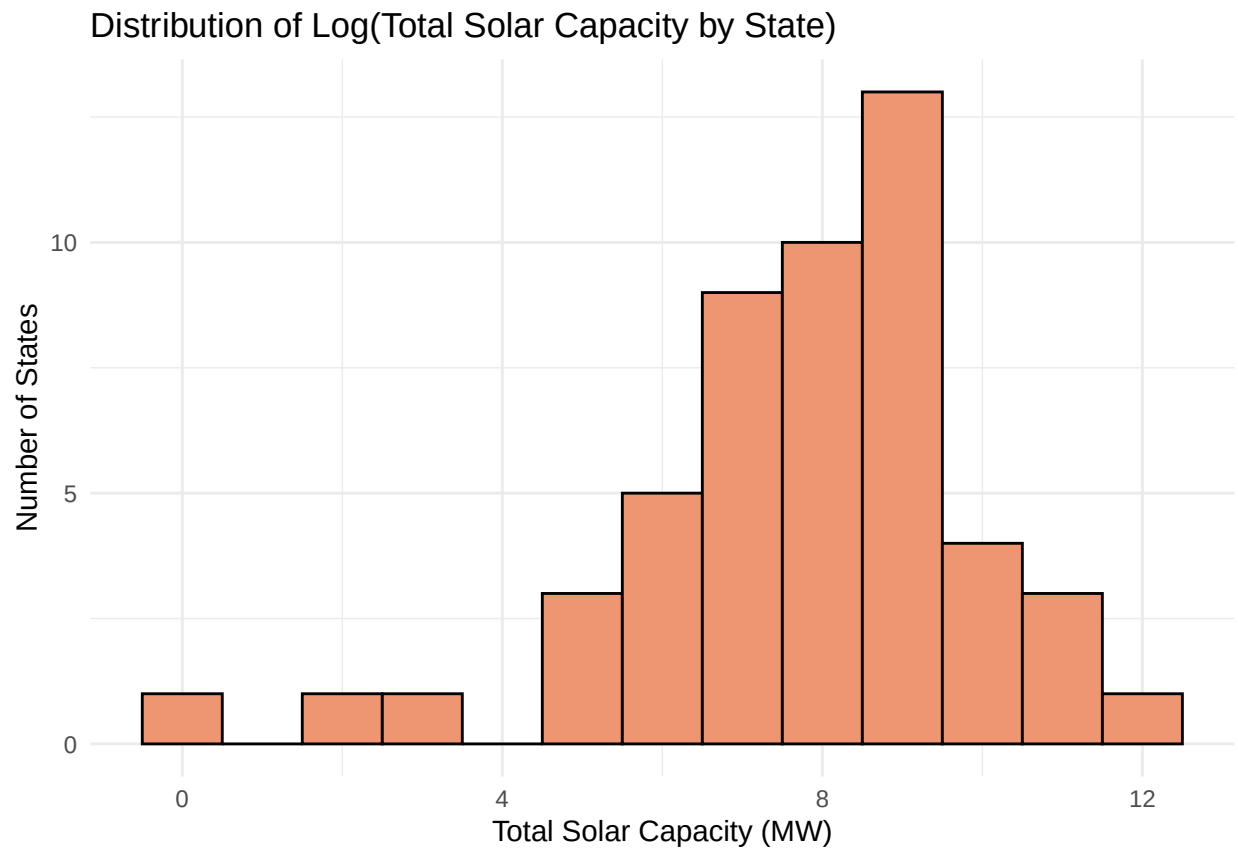
# Sum total capacity per state across all years
solar_total_state <- solar_all_years %>%
  group_by(State) %>%
  summarise(TotalCapacity = sum(TotalCapacity, na.rm = TRUE))

wind_total_state <- wind_all_years %>%

```

```
group_by(State) %>%
  summarise(TotalCapacity = sum(TotalCapacity, na.rm = TRUE))
```

```
ggplot(solar_total_state, aes(x = log(TotalCapacity))) +
  geom_histogram(binwidth = 1, fill = "lightsalmon2", color = "black") +
  labs(
    title = "Distribution of Log(Total Solar Capacity by State)",
    x = "Total Solar Capacity (MW)",
    y = "Number of States"
  ) +
  theme_minimal()
```



```
ggplot(wind_total_state, aes(x = log(TotalCapacity))) +
  geom_histogram(binwidth = 1, fill = "darkblue", color = "black") +
  labs(
    title = "Distribution of Log(Total Wind Capacity by State)",
    x = "Total Solar Capacity (MW)",
    y = "Number of States"
  ) +
  theme_minimal()
```

